

Monday, July 27, 2026

10:00 a.m. – General Discussion

Commissioners Ron Smith, Chair; Colene Martin, Vice-Chair; Gary Richardson; Nikole Guzman, Recorder

Additional Attendees: Ryan Johnson, Facilities Director; Scott Hyde, Community Justice Director; Dave Streeter, Forestry Director; Sheriff Dave Daniel; Rob Brandes, Public Works Director; and Matt Weinberger, Day Wireless Systems

Chair Ron Smith called the meeting to order at 10:00 a.m.

1. Declaration of Potential Conflicts of Interest

All Commissioners reviewed today's agenda, and it was determined that there are no conflicts of interest.

2. Communications

Commissioner Richardson explained that the purpose of the meeting was to identify the radio communications needs of the Sheriff's Office and other County departments and develop a coordinated plan for replacement and modernization of the County's public safety communications system. The Board invited department representatives to describe their operational needs, known coverage deficiencies, and concerns regarding the current system.

Current Radio System and Infrastructure

Facilities Director Ryan Johnson reported that the existing communications system is obsolete, manufacturers no longer support much of the equipment, and replacement parts are increasingly difficult or impossible to obtain. The current system consists primarily of four mountaintop sites and equipment located at the Justice Building, connected through a microwave network. The system supports separate channels used by the Sheriff's Office, Public Works, general government, and other County operations.

Federal Engineering was completing an assessment of the system, including existing coverage and potential locations for additional sites. Staff anticipated that the assessment would address significant deficiencies in the Williams and Cave Junction areas, where radio service is limited or nonexistent in some locations. The Board was awaiting the final report and associated coverage maps.

Commissioner Martin reported that she had contacted John Lee, Oregon's Statewide Interoperability Coordinator, regarding the County's communications challenges. His office supports interoperable and resilient emergency communications throughout Oregon and may assist the County in positioning the project for Homeland Security and other federal grant opportunities. Mr. Lee offered to review communications sites throughout Josephine County and identify existing federal, state, local, or privately owned towers that might support the County's system.

Communication Site Options

Matt Weinberger of Day Wireless Systems explained that the County could either purchase property and construct its own towers and equipment shelters or lease space at existing communications sites. County-owned facilities would require a larger initial investment but would provide the County with direct control over the site and its ongoing maintenance.

Leasing or co-locating at an existing tower could reduce some construction costs, particularly when power and other utilities are already available. However, a shared site must be evaluated for radio-frequency interference. Sites already carrying VHF communications may require complex filters and signal-combining equipment, which can increase cost and reduce system performance. Day Wireless recommended identifying technically compatible sites that are not already heavily occupied by similar VHF systems.

Mr. Weinberger explained that the County's existing VHF frequencies are closely grouped, requiring extensive filtering. Filtering reduces the amount of radio signal ultimately transmitted from the antenna. The County's equipment was designed around older wideband standards and has struggled to maintain coverage following the federal narrow banding requirements implemented in 2013.

VHF and Supplemental Technologies

Commissioners asked whether the County should move to a completely different radio technology. Mr. Weinberger explained that VHF remains well suited to Josephine County because it travels farther with fewer sites and performs comparatively well through mountainous and heavily forested terrain. Higher-frequency UHF and 700/800 MHz systems may perform better inside buildings and dense urban areas but generally require more towers to cover the same rural geographic area.

Day Wireless recommended maintaining a County-controlled VHF system as the primary communications network. Supplemental technologies, including satellite service, Starlink, higher-frequency radios, fiber connections, and building-specific systems, could then be added to improve coverage in isolated areas or inside facilities such as the Jail and hospitals. Mr. Weinberger cautioned against relying exclusively upon outside satellite or commercial services because the County would not control those systems or their availability during an emergency.

Preliminary Replacement Cost

Mr. Weinberger reviewed a 2022 proposal developed by Motorola and Day Wireless. At that time, replacement of the primary system was estimated at approximately \$3.3 million, with roughly \$800,000 in optional lifecycle and system-upgrade services. He estimated that a comparable project would likely cost approximately \$5 million or more in 2026, before adding new mountaintop sites or other major system expansions.

The optional lifecycle agreement would allow Motorola and Day Wireless to inspect the system approximately every two years, replace equipment that had reached the end of its supported life, and install necessary software and firmware updates. Commissioners discussed the importance of maintaining the future system rather than allowing it to remain unchanged until it again became obsolete.

Mr. Weinberger also advised that the County's dispatch console and core equipment had fallen behind current standards. Motorola had released a newer console platform that could be incorporated into the replacement project. Promotional pricing was temporarily available on certain console positions, but Commissioners agreed that the project should be designed around long-term operational needs rather than a short-term discount.

Power and System Reliability

Facilities staff reported that several recent failures involved power systems at remote communications sites. The Onion Mountain site relies primarily upon solar power, batteries, and a generator. Staff was replacing and strengthening those systems, although bringing commercial electrical service to the site could be prohibitively expensive because the nearest utility infrastructure may be several miles away.

Day Wireless described larger solar and battery systems that could provide more reliable long-term power at remote sites. A properly designed direct-current power plant could include expanded solar panels, multiple battery banks, inverters, converters, and a more dependable generator. Although such systems represent a significant investment, they may cost substantially less than extending commercial electrical service over several miles.

The Board discussed the need to design the system for severe weather, including heavy rain, snow, extended power outages, and wildfire emergencies. Day Wireless recommended strengthening backup power at each site and creating multiple communications paths so that a failure of one connection would not disable the entire system.

Fiber could serve as the primary connection at sites where it is available, with microwave or satellite service retained as a backup. Day Wireless emphasized that no single connection should be treated as completely reliable and that the County should maintain redundant paths wherever practical.

Commissioners also supported regularly testing the system under failover conditions rather than assuming backup equipment would function during an emergency. Day Wireless recommended incorporating communications failures into mass-casualty and emergency-response exercises involving law enforcement, Emergency Management, Public Works, medical providers, and other participating agencies.

Community Justice Communication Needs

Community Justice Director Scott Hyde reported that probation officers and work crews regularly travel throughout Josephine County. His department had tracked locations where employees experienced weak or unreliable radio service and provided that information to Facilities and Public Works.

Community Justice staff travel throughout the Illinois Valley, Williams, Wolf Creek, the Speaker Road area, communities near the Jackson County border, and other remote portions of the County. Mr. Hyde described locations where radio service disappears after employees travel behind hills or into valleys. He acknowledged that covering every isolated location may not be financially practical but emphasized the need to obtain the greatest possible coverage for employees working alone or transporting work crews.

Sheriff's Office Communication Needs

Sheriff Dave Daniel stated that the communication deficiencies have directly affected Sheriff's Office operations and deputy safety. Deputies are familiar with known dead zones and must modify how they respond or deploy personnel in areas where radio service is unreliable.

The Sheriff characterized the situation as serious and supported both restoration of the existing system for immediate needs and development of a comprehensive replacement plan. He expressed appreciation that County departments, Facilities, Day Wireless, and the Board were working together to address the problem.

Public Works Communication Needs

Public Works Director Rob Brandes reported that road crews experience many of the same dead zones identified by Community Justice and the Sheriff's Office. Page Mountain had historically presented a significant problem, and communications can deteriorate during winter storms and other severe weather.

Public Works often sends employees into remote areas in pairs, but radio communication remains important when crews are responding to landslides, fallen trees, snow, flooding, and road closures. The Board emphasized that the system should be designed to function during those severe conditions because that is when reliable communication is most critical.

Day Wireless discussed equipment capable of allowing an employee carrying a low-power handheld radio to remotely use the higher-powered mobile radio installed in a nearby County vehicle. Such equipment could extend an employee's effective range while working away from the vehicle.

Forestry and Parks Communication Needs

Forestry Director Dave Streeter explained that Forestry uses the County radio system primarily for vehicle-to-vehicle communication during special projects and to monitor Oregon Department of Forestry radio traffic during wildfire season. The department currently relies heavily upon cell phones because the cost previously proposed for full participation in the County radio system exceeded the benefit Forestry expected to receive.

Mr. Streeter stated that Forestry would participate if directed and noted that the department manages County land that might provide suitable locations for future communications sites. Once coverage maps identified preferred locations, Forestry could assist in determining whether the County already owned appropriately situated property.

Commissioner Martin stated that Parks should also be considered during system planning. Parks employees travel to remote facilities where cellular service may be unreliable, including Whitehorse Park and parks along the Rogue River. Although Parks was not currently a significant radio-system user, the department could benefit from an emergency communication option if the cost were reasonable.

Emergency Management Participation

The Board discussed the need to include the County's Emergency Management Director in the project. Commissioner Martin reported that the new Director had experience with a recent radio-system upgrade in another state and had expressed interest in participating in the planning process.

Commissioners agreed that Emergency Management should help evaluate site resiliency, backup power, mass-casualty exercises, wildfire communications, and coordination among participating agencies.

Interoperability and P25 Standards

Mr. Weinberger explained that federal funding generally requires new public safety radio equipment to comply with Project 25, commonly referred to as P25. P25 establishes technical standards allowing digital public safety communications equipment from different manufacturers and agencies to operate together.

The County could continue operating conventional VHF channels while purchasing equipment capable of future P25 digital operation. The Sheriff's Office already possessed many radios capable of P25 communication, although the equipment would need to be inventoried, programmed, and activated.

The Board discussed the advantages and disadvantages of analog and digital communication. Digital radio can provide clearer sound and extend useful coverage in some areas, but the signal may disappear abruptly when it becomes too weak. An analog signal may remain partially understandable through static in fringe areas. Day Wireless recommended designing a system capable of supporting both modes, with analog retained where appropriate and digital channels added for law enforcement and other specialized operations.

Day Wireless also recommended considering encryption for certain law-enforcement communications because of evolving security expectations concerning protected information transmitted over public radio systems.

Cave Junction and Additional Sites

The Board identified improved coverage in Cave Junction, Williams, and the Illinois Valley as a priority. Day Wireless noted that Cave Junction previously had a communications site and that the County retains a microwave license directed toward that area. Once a technically appropriate site is identified, the County could revise the microwave licensing and add Cave Junction to the replacement project.

Commissioners agreed that the location should be selected based upon engineering and coverage requirements rather than simply using County-owned property because it was available. However, County land that provided nearly equivalent coverage could reduce acquisition or leasing costs and should be carefully evaluated.

Fiber and Redundant Connections

Commissioner Richardson asked whether fiber could replace some of the microwave links installed more than 20 years ago. Day Wireless stated that fiber is increasingly used as the primary connection when available, while microwave remains as a secondary or backup path.

Staff and Day Wireless agreed to contact local fiber and utility providers, including AT&T, Hunter Communications, and potentially utility or governmental broadband networks, to determine where service might be available near existing or proposed communications sites. The Onion Mountain location was identified as one of the most difficult sites because the County leases the property and commercial power and fiber may be distant.

Funding and Financing Options

Commissioner Martin reported that she was pursuing federal appropriations and other grant opportunities through the offices of Oregon's congressional delegation. Day Wireless explained that federal appropriations may reimburse certain eligible expenses already incurred, allowing the County to begin a project while continuing to seek outside funding.

Motorola also offers financing that may extend up to 15 years, with the possibility of deferring the first payment for approximately one year. Under such an arrangement, the County could complete engineering, order the equipment, install the system, and potentially begin operating before the first annual payment became due.

Commissioners discussed Homeland Security grants, federal appropriations, financing, County reserves, and possible interfund loans. They agreed that grant opportunities should be aggressively pursued but that the County could not postpone action indefinitely while waiting for outside funding because the existing system was in a dangerous and increasingly unstable condition.

Commissioner Richardson discussed spreading the cost of a long-lived communications system over a period of 20 or 30 years rather than requiring current taxpayers to bear the entire cost immediately. He clarified that he was not proposing a new tax. Financing would instead allow the County to use existing revenues over time while distributing the cost among the taxpayers who would benefit from the system during its useful life.

Procurement and Project Timing

Day Wireless explained that Motorola equipment may be purchased through the National Association of State Procurement Officials cooperative contract, commonly referred to as NASPO. The cooperative contract provides competitively procured pricing and may satisfy applicable public contracting requirements. Day Wireless also participates in another Oregon governmental purchasing contract that could apply to portions of the work. Legal Counsel would be consulted before the Board selected a procurement method.

A preliminary rough-order-of-magnitude estimate could potentially be prepared within several weeks, although a more detailed and binding proposal could require 60 to 90 days. Manufacturing, delivery, site construction, and installation would require additional time after a contract was executed. Day Wireless advised that a formal Board expression identifying the project as a priority could help move the County higher in Motorola's engineering and proposal schedule.

Scope and Long-Term System Design

Commissioners requested that Day Wireless develop a complete project framework identifying essential components, strongly recommended improvements, and optional future enhancements. The Board wanted to understand the complete system before reducing or phasing the project based upon cost.

Day Wireless agreed to prepare a prioritized list that would distinguish the fundamental replacement system from supplemental options such as satellite-capable radios, additional communications sites, expanded digital functions, upgraded power systems, fiber connections, and other enhancements. Mr. Weinberger cautioned that preparing numerous detailed pricing alternatives could delay the project because each technical choice affects several other components.

The Board asked Day Wireless to describe a realistic, durable system that would provide the greatest coverage and reliability without attempting to replicate the substantially larger and more expensive regional system constructed in Jackson County. Mr. Weinberger explained that a fully trunked regional system involving numerous agencies could cost tens of millions of dollars and was not an appropriate direct comparison for Josephine County.

A modern conventional VHF system could provide the County with a reliable base at a significantly lower cost. The core equipment could be designed so that simulcast, digital operation, trunking, additional sites, and other improvements could be added in later phases.

Commissioners supported constructing a strong, dependable base system with the technical capability to accommodate future improvements. The system was compared to a modular structure in which the essential foundation would be completed first and additional components could be added over the next five, ten, or fifteen years without replacing the underlying infrastructure.

Next Steps

Day Wireless agreed to coordinate with Motorola Engineering, Federal Engineering, Facilities, and the participating County departments to prepare a project scope and preliminary cost estimate. Day Wireless would also conduct a more detailed review of the County's remote sites with its construction personnel to evaluate towers, shelters, generators, solar panels, batteries, and other power requirements. The Emergency Management Director would be invited to participate in the site review.

The Board requested that the initial scope include the County's essential replacement needs and show how the system could be expanded over time. Commissioners stated that they wanted to understand the entire project before selecting individual phases or removing components solely to reduce the initial estimate.

The Board agreed that reliable emergency communications were among the County's highest operational priorities. No formal expenditure or procurement decision was made. The matter will return for additional discussion after the Federal Engineering report, Motorola information, project scope, and preliminary estimates are available. A specific follow-up meeting date was not established.

3. Other

4. Additional Subject Matters from Commissioners/Announcements

The meeting adjourned at 11:31 a.m.